

Francisco García Atienza

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Mathematician and Programmer specializing in Data Science and SciML

Career Objectives

I am a mathematician and computer programmer with a background in numerical analysis and SciML. My main interests lie at the intersection of mathematics and programming, with a strong focus on data science and applied AI/ML. Combining solid analytical skills with practical programming experience, I am motivated to contribute to data-driven projects where predictive modeling and AI-driven solutions advance scientific discovery.

Education

- 2023-2025 **MSC IN NUMERICAL ANALYSIS** *Lund University (Sweden)*
Thesis title: "Training Robust Reduced-Order Models using the Adjoint Method"
lup.lub.lu.se/student-papers/record/9200627
Developed numerical techniques for constructing stable and efficient reduced-order models, leveraging adjoint-based optimization to enhance robustness in data-driven settings.
- 2020-2023 **BSC IN MATHEMATICS** *Lund University (Sweden)*
Thesis title: "A Continuous-Time ODE Framework to Centrality in Dynamic Networks"
lup.lub.lu.se/student-papers/record/9120052
Explored novel mathematical model for measuring centrality in evolving networks through a continuous-time ODE approach in order to analyze dynamic interactions.
- 2010-2013 **BSC IN COMPUTER SCIENCE** *Universidad Politécnica de Valencia (Spain)*
Focused on programming, algorithms, and software development with practical experience in languages such as Python, Java, and R. Gained a strong foundation in data structures, databases, web technologies, and cloud-based platforms, as well as problem-solving and analytical skills applied through academic projects and coursework

Work

- 2018–2020 **JUNIOR DATA SCIENTIST** *ScanOats - Lund University (Sweden)*
Developed data analysis and predictive modeling workflows for biological datasets:
- Applied statistical and machine learning methods to uncover patterns and support predictive biological insights.
 - Developed automated workflows and visualization tools to facilitate data interpretation for research teams.
- 2015-2017 **JUNIOR DATA ANALYST** *Verisure (Sverige)*
Python and R programming on:
- Collected, cleaned, and processed large datasets to extract meaningful insights.
 - Developed statistical models and predictive analytics to identify trends and business opportunities.
 - Automated data reporting processes.
- 2014-2015 **JUNIOR FULL-STACK DEVELOPER** *Umeå Plant Science Centre (Sweden)*
Project management and designing of bridge components.
- **WebApollo Project:** Contributed to the development of an annotation platform (apollo.berkeleybop.org). Developed and maintained genome visualization tools. Various functionalities were implemented using JavaScript, PHP, and Python in a MySQL database to manage tree population data.

Courses

2025	DATA VISUALIZATION <i>Linnæus University (Sweden)</i> Data visualization techniques using R and Python. Transformation of complex data into actionable insights, enhancing communication across diverse audiences for effective decision-making.
2024	NUMERICAL ALGORITHMS WITH PYTHON/SCIPY <i>Lund University (Sweden)</i> SciPy/NumPy data structures, object-oriented programming, and integration with C/Fortran libraries. Automated testing for computational methods.
2024	IMAGE ANALYSIS <i>Lund University (Sweden)</i> Fundamental techniques in digital image analysis with applications in computer vision, multispectral analysis, and statistical image analysis.
2023	APPLIED AI <i>Luleå University (Sweden)</i> Supervised and unsupervised learning, neural networks, decision trees, and reinforcement learning. Practical implementation focusing on model training, evaluation, and optimization.
2023	SOFTWARE DEVELOPMENT WITH JAVA <i>Luleå University (Sweden)</i> Object-oriented programming in Java, including data structures, algorithms, and software development with practical applications.

Software Skills

- **Programming & Scripting:** Python (pandas, NumPy, matplotlib, seaborn, scikit-learn, PyTorch), R, Bash
- **Machine Learning:** Supervised & unsupervised learning (linear and logistic regression, decision trees, random forests, clustering, PCA), cross-validation, model evaluation metrics (accuracy, F1-score, RMSE)
- **Statistics & Math:** Descriptive statistics, hypothesis testing, linear algebra, regression models, Monte Carlo / Markov Chain methods (MCMC)
- **Numerical Analysis & Scientific Computing:** SciPy, MATLAB toolboxes
- **Version Control & Collaboration:** Git, GitHub Actions, virtual environments (conda/pip)
- **Data Manipulation & Management:** Data cleaning, preprocessing, feature engineering, handling structured & semi-structured data
- **Databases:** PostgreSQL, MySQL, MariaDB, MongoDB, Excel
- **Visualization & Reporting:** ggplot2, Plotly, Jupyter, RMarkdown, Power BI, Tableau, $\text{\LaTeX}$
- **DevOps & Cloud:** Azure, GCP, basic DevOps / CI/CD for data pipelines, Docker and Kubernetes
- **Agile Methodologies:** Scrum (practical experience with sprints, backlog management, daily stand-ups, and retrospectives)
- **Other Technologies:** Java, macOS/Linux/Windows administration, Apache Web Server, CMS

Personal Profile

GitHub Portfolio	https://fragarat.github.io
Residence	Malmö, Sweden (open to relocate within Sweden) and Torrevieja, Spain
Nationality	Spanish and Swedish
Driving license	B

Languages

Spanish	Mother tongue
English	Advanced, C1
Swedish	Intermediate, B2 (Komvux SAS3)

References

Mengwu Guo	mengwu.guo@math.lu.se
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Philipp Birken	philipp.birken@math.lu.se